PHYS 661 Fall 2024

Problem set 2, due 9/19 2024. Each problem is 10 pts unless otherwise stated. Submit your homework either (1) via Brightspace or by (2) emailing it directly to the TA at gbowling@purdue.edu.

1 N-state all-to-all system (20 pts)

Consider an N-state system with a perturbation

$$V = \begin{pmatrix} v & v & \dots & v \\ v & v & & \\ \vdots & & \ddots & \\ v & & & v \end{pmatrix}, \quad v \in \mathbb{R}, \tag{1}$$

i.e., $\langle n^{(0)}|V|m^{(0)}\rangle = v$ for all n, m. Let us for simplicity assume that the unperturbed Hamiltonian is just zero, $H_0 = 0$, i.e. $E_n^{(0)} = 0$ for all $n = 1, 2, \dots, N$. In this case we can use Brillouin-Wigner (BW) perturbation theory.

1. (5 pts) Before doing perturbation theory, find the eigenvalues of V and their degeneracies exactly.

2. (5 pts) By using BW perturbation theory, evaluate the correction to the energy E of a state up to and including third-order corrections in v. You should get an equation for E of the form $E = a_0 + a_1 v + a_2 \frac{v^2}{E} + a_3 \frac{v^3}{E^2}$. (No need to try to solve that equation.)

3. (10 pts) Going beyond 3rd order, the equation for the perturbed energy E will be of the form $E = a_0 + \sum_{k=0}^{\infty} a_{k+1} v (\frac{v}{E})^k$. Find the general nth order coefficient a_n. Using your result, sum up the series $\sum_{k=0}^{\infty} a_{k+1} v (\frac{v}{E})^k$. Solve the resulting equation for E.

2 1D δ-potential well under electric field (10 pts)

Consider a 1D potential well $U(x) = -\alpha\delta(x)$. The ground state of this system is a bound level with wave function $\psi_0(x) = \sqrt{\kappa} e^{-\kappa|x|}$ and energy $E_0^{(0)} = -\frac{\hbar^2\kappa^2}{2m}$, where $\kappa = m\alpha/\hbar^2$. The excited state wave functions are $\psi_{+,k}(x) = \frac{1}{\sqrt{\pi}} \cos(k|x| + \frac{\varphi_k}{2})$ and $\psi_{-,k} = \frac{1}{\sqrt{\pi}} \sin kx$, both with energy $E_k^{(0)} = \frac{\hbar^2 k^2}{2m}$. (The subscript $\pm$ denotes spatial parity eigenvalues.) Consider a perturbation $V = bx$ acting on the ground state. Calculate the first and second order energy shifts to the ground state energy. In the case of continuous spectrum the 2nd order correction to the ground state energy is

$$E^{(2)} = \sum_{\sigma=\pm} \int_0^\infty dk \frac{1}{E_0^{(0)} - E_k^{(0)}} |\langle \sigma, k|V|0\rangle|^2 . \tag{2}$$

Hint: $\int_0^\infty dx x e^{-\kappa x} \sin kx = \frac{2\kappa k}{(\kappa^2+k^2)^2}$.